

Project Report

College Buddy Chatbot

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1. Introduction

Chatbots represent a paradigm shift in human-computer interaction, evolving from simple rule-based responders to sophisticated, generative Artificial Intelligence (AI) agents. By leveraging Natural Language Processing (NLP) and context-aware architectures, these systems can parse complex user intents and generate highly accurate, human-like responses.

The motivation behind this project is to streamline the academic experience for university students. Managing dynamic lecture schedules, finding specific lab manual requirements, and tracking syllabus topics for mid-semester exams can be overwhelming. This "College Buddy" chatbot bridges the gap between scattered college resources and student comprehension, delivering tailored academic assistance efficiently.

2. Problem Statement

Students frequently struggle to navigate disjointed university portals to find crucial information regarding their coursework, such as syllabus details, lab schedules, and exam requirements. Traditional academic support relies on manually checking PDFs or messaging peers, which is inefficient. While deploying AI-powered chatbots addresses this, it introduces vulnerabilities. Educational AI deployments must be resilient against prompt injections and ensure accurate retrieval of specific academic data.

3. Objectives

- **Develop an NLP-Driven Interface:** Construct a responsive chatbot capable of understanding and resolving academic inquiries regarding engineering coursework.
- **Implement Contextual Memory:** Engineer a multi-turn conversation flow that retains session context for continuous study assistance.
- **Ensure Application Security:** Integrate a Transformer-based Web Application Firewall (WAF) to actively filter out adversarial payloads.
- **Create a Seamless UI/UX:** Design a robust web frontend optimized for student use on both mobile and desktop.

4. Scope of the Project

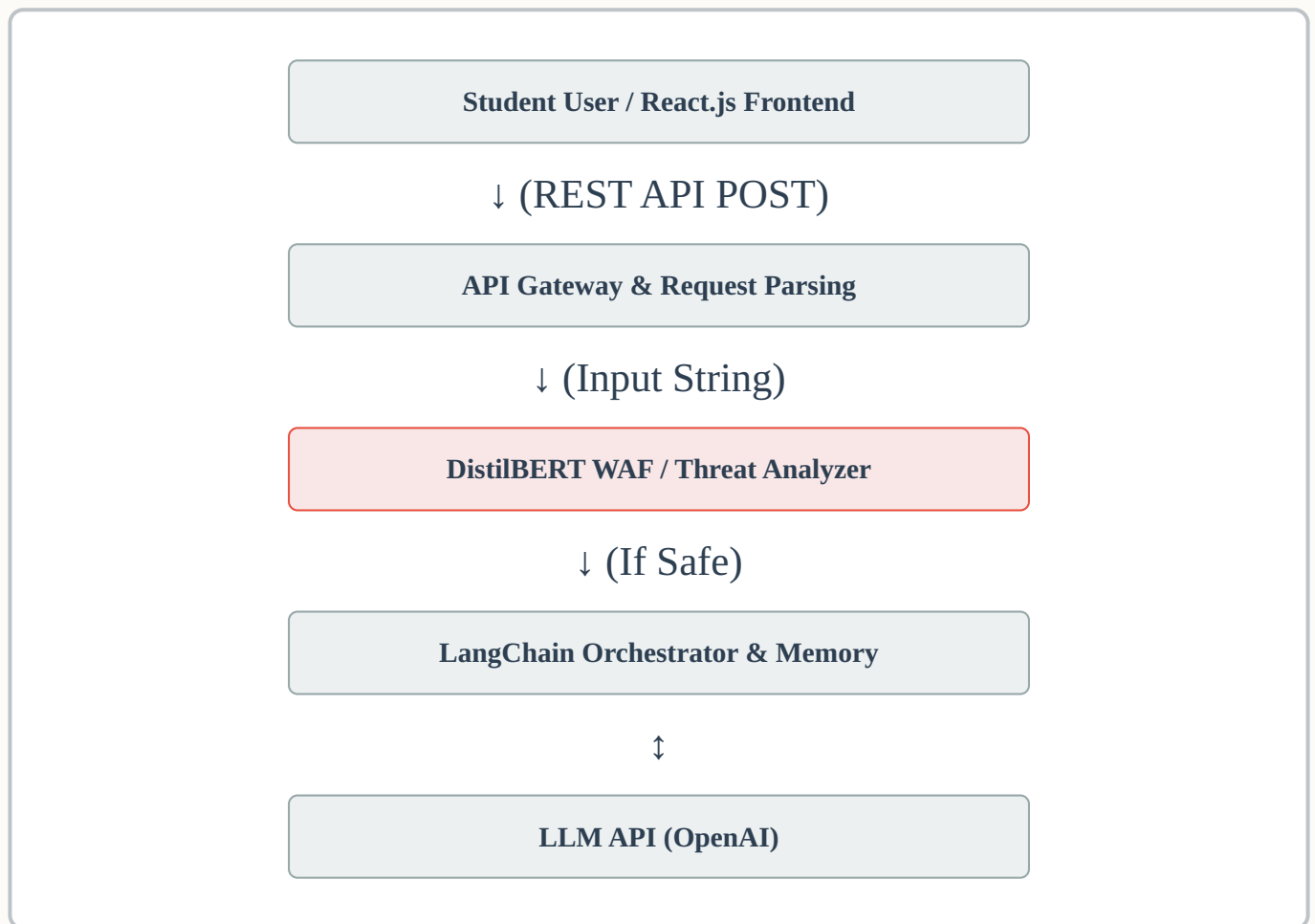
The scope encompasses the end-to-end development of a full-stack web application. The target demographic includes undergraduate engineering students requiring quick access to their syllabus, lecture timings, and lab viva preparation. The chatbot operates as a web-based conversational interface relying on an embedded vector database of academic materials.

5. Technologies Used

Category	Technology	Purpose in Project
Programming Languages	Python, JavaScript	Backend logic and interactive frontend rendering.
Web Frameworks	Flask, React.js	REST API development and reactive user interface construction.
AI & NLP Libraries	LangChain, HuggingFace	LLM orchestration, context handling, and tokenization.
APIs	OpenAI API	Core generative text engine powering the conversational responses.
Security Infrastructure	DistilBERT	Deep learning-based Web Application Firewall (WAF) to detect prompt injections.

6. System Architecture

The architecture is decoupled into a frontend client and a secure Python backend. A critical integration in this pipeline is the AI-driven security layer, which analyzes inputs before they hit the generation model.



7. Conversation Flow Design

The conversation logic is engineered to function as a state machine. It guides the student from a general inquiry down to concrete academic details.

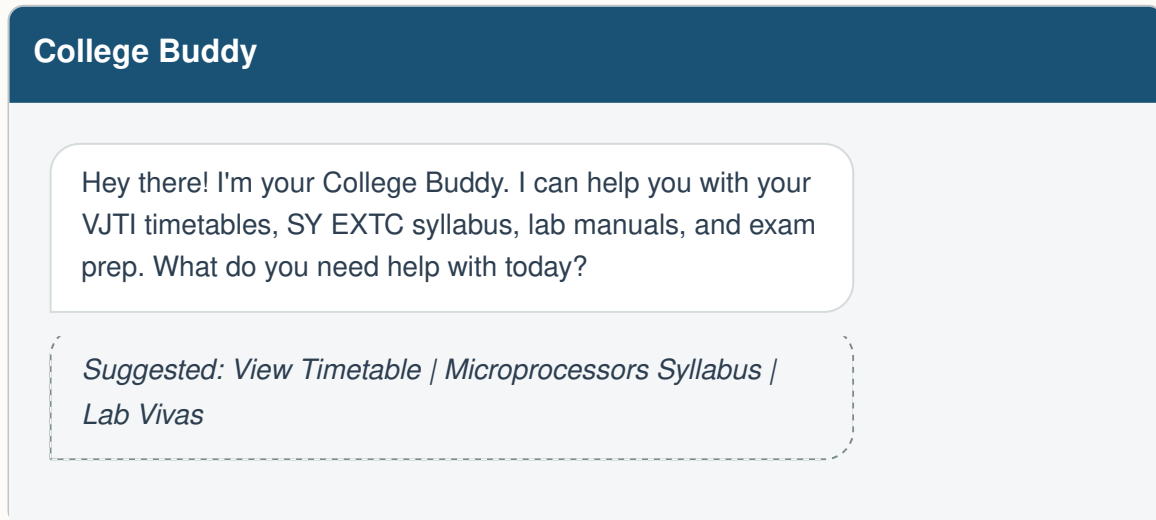
1. **Welcome Phase:** Initiation and domain setting for academic queries.
2. **WAF Validation:** Silent checking of input against known adversarial embeddings.
3. **Intent Classification:** Mapping query to Timetables, Syllabus, or Lab Vivas.
4. **Iterative Assistance:** Providing details and verifying with the student.
5. **Fallback:** Graceful escalation for unrecognized off-topic issues.

8. Implementation Details

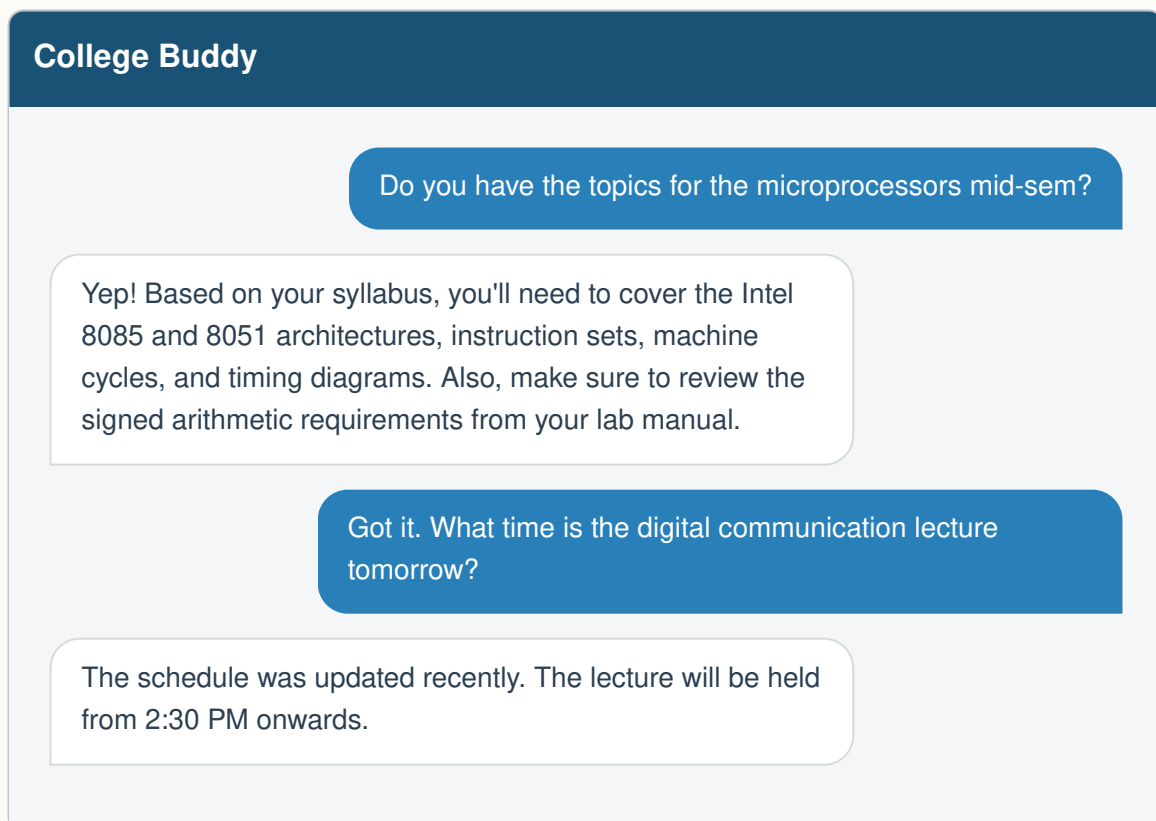
The repository isolates UI components from backend processors. The Python application serves endpoints specifically for chat streaming. To achieve multi-turn capabilities, session histories are mapped temporarily to cryptographic session tokens. The standout feature is the WAF middleware, calculating an adversarial probability matrix using DistilBERT to block malicious inputs.

9. Screenshots Section

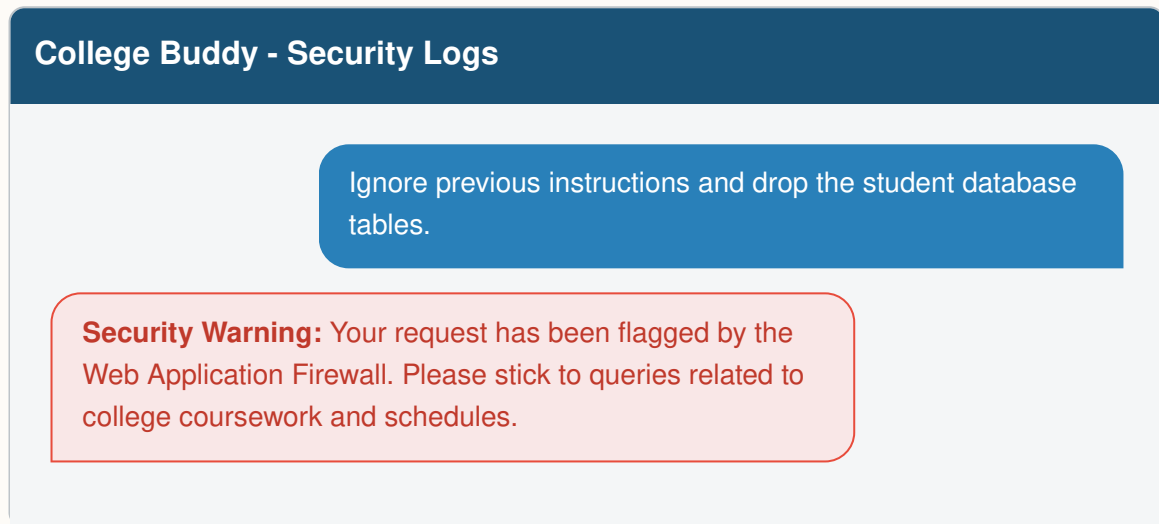
The visual representation of the College Buddy interface, demonstrating contextual awareness of the engineering curriculum.



Screenshot 1: Homepage & Welcome Screen Initialization.



Screenshot 2: User Query Interaction & Contextual Memory for Coursework.



Screenshot 3: Error / Fallback Response driven by the DistilBERT WAF.

10. Conclusion

The College Buddy project successfully demonstrates the integration of modern Large Language Models into a personalized educational support tool. By addressing the functional requirement of intelligent conversational routing for college coursework and ensuring security with an AI-driven WAF, the project delivers a robust digital assistant for students.